

TransEQ

A hybrid program equaliser: passive-style curves at the ends, versatile bells in the middle, and transformer iron on the way in and out.



Overview

TransEQ combines the broad, musical curves of a classic passive program equaliser at the low and high ends with two more flexible bell bands in the middle. The signal passes through an input transformer, the EQ and an output transformer.

The low band has the famous trick of the passive designs: BOOST and ATTEN are separate and can be used together. The boost is a slightly resonant shelf and the cut a much broader one, so using both lifts the lowest lows while carving a dip above them. The high band has a boost bell with its own BANDWIDTH and a separate cut shelf.

The IRON transformers saturate on magnetic flux, so the low end thickens first with mostly low-order harmonics while the mids and highs stay clean; below saturation they are transparent. A very slight drift is built in: the band frequencies and gains wander a little, like components warming up. The name is a tribute to the trans community, hence the blue and pink accents.

Reach for it for

- Broad, musical tone shaping on a mix bus or master
- The classic 'boost and cut' low end on kick, bass and full mixes

- Air and presence on vocals and acoustic instruments
- Adding transformer weight and density to the lows
- Quick clean-up with the HPF and LPF

Controls

Big knobs: drag up and down (Shift for fine control) or use the mouse wheel; double-click to reset. The value shows while you hover. Stepped selectors: click a printed value to pick it, or click, drag or scroll on the knob to step through.

The header

S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.
EQ IN	Switches the EQ curves in and out (in by default; the light shows when it is in). The transformers stay in circuit either way, so you can compare the EQ alone. While it is out the EQ section is dimmed.

LOW

FREQ	The low band's frequency: 20, 30, 60, 100 or 150 Hz.
BOOST	A slightly resonant low shelf, 0 to 10 (up to about +14 dB).
ATTEN	A much broader, gentler low cut, 0 to 10 (up to about -13 dB). Used together with BOOST it lifts the lowest lows with a dip above them - at 60 Hz with both full, about +7 dB at 30 Hz and -6 dB at 100 Hz.

LO MID and HI MID

Frequency selector	LO MID: 150, 220, 330, 470, 680 Hz, 1 kHz or 1.5 kHz. HI MID: 1, 1.5, 2.2, 3.3, 4.7, 6.8 or 10 kHz.
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WIDE / NARROW	The base width of the bell. Click to switch.
GAIN	Boost or cut, -12 to +12 dB. The bell narrows as the gain goes up (proportional Q), so small moves are broad and gentle, big ones more focused.

HIGH

BOOST FREQ	The high boost's frequency: 3, 4, 5, 8, 10, 12 or 16 kHz.
ATTEN FREQ	Where the high cut starts: 5, 10 or 20 kHz.
BOOST	A high bell, 0 to 10 (up to about +16 dB).
BANDWIDTH	The width of the high boost, 0 (sharp) to 10 (broad). 5 by default.
ATTEN	A high shelf cut, 0 to 10 (up to about -16 dB).

Bottom row

HPF (yellow cap)	A 12 dB / octave high-pass, 20 - 500 Hz. Fully left is OFF (out of circuit).
LPF (black cap)	A 12 dB / octave low-pass, 2 - 20 kHz. Fully right is OFF.
IRON IN	How hard the input transformer is driven, 0 to 10 (3 by default). Higher settings thicken the lows with mostly low-order harmonics; the mids and highs stay clean.
IRON OUT	The same for the output transformer, after the EQ and OUTPUT. Boosting the lows into it saturates it more.
OUTPUT	Output level, -12 to +12 dB, set before the output transformer - so it also changes how hard IRON OUT works.
LEVEL	LED meter of the output level.

Two stages work in the background with no controls on the panel: an input guard that gently eases very hot sources down towards about -18 dBFS average and

catches fast peaks above -6 dBFS (it never boosts quiet sources), and a soft clipper that holds the output peaks at -0.3 dBFS.

Tips

- For the classic low end, set FREQ to 30 or 60 Hz and turn BOOST and ATTEN up together: the lows get bigger and tighter at the same time.
- For air without harshness, use HIGH BOOST at 10 - 16 kHz with a broad BANDWIDTH, and a touch of ATTEN at 20 kHz if it gets fizzy.
- Use EQ IN to compare: the transformers stay in, so you only hear what the EQ curves are doing.
- Turn IRON IN and IRON OUT to 0 for a clean EQ; raise them on bass, kick and mix bus for weight.
- Because OUTPUT sits before the output transformer, raising it pushes IRON OUT harder - lower IRON OUT if you only want more level.

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